Kanghwi Lee

Audio & Speech ML | Foundation Model Adaptation & Generative Modeling
Seoul, South Korea | Zurich, Switzerland
kanlee@ini.ethz.ch | Website | GitHub | Google Scholar | LinkedIn

Profile

Audio and speech ML researcher and Ph.D. candidate at ETH Zurich with 4+ years of experience developing models and evaluation workflows for large-scale audio datasets. Currently researching TTS audio watermarking at NAVER Cloud. Work spans Whisper-based voice activity detection, generative modeling over learned audio representations, and scalable audio clustering across 80,000+ hours of recordings. Strong Python/PyTorch background; Ph.D. completion expected early 2027.

Professional Experience

NAVER Cloud Jul 2026–Present
Research Intern – TTS Audio Watermarking, Voice Synthesis Team

- Researching TTS audio watermarking for scalable synthetic-speech provenance, targeting high-volume TTS service settings.

ETH Zurich, Institute of Neuroinformatics Apr 2022–Present
Doctoral Researcher – Audio ML & Computational Neuroscience

- Developing audio ML methods for large-scale audio datasets, from scalable segmentation and clustering pipelines to learned representations, generative modeling, and quantitative evaluation.
- Doctoral research focuses on how juvenile songbirds form and update vocal templates, combining large-scale audio analysis with computational modeling of vocal learning.

Selected Speech & Audio ML Research

Trajectory Variance – Counterfactual Generative Modeling Interspeech 2026 Oral | arXiv | code
PyTorch, VAEs, Learned Audio Representations, Optimal Transport, Counterfactual Generation

- Developed Trajectory Variance, a label-free framework for estimating how individual audio events would change across developmental time, and evaluated it on approximately 680K vocalizations.
- Built an age-conditioned displacement model for single-pass latent-shift prediction with mini-batch optimal-transport pairing; also prototyped a Flow Matching formulation before selecting the displacement approach.
- Across all tested birds, trajectory variance was consistently higher for learned song syllables than innate calls after controlling for duration (Cohen's $d = 0.29$–0.57; AUC = 0.58–0.67).

WhisperSeg – Whisper Adaptation for Voice Activity Detection ICASSP 2024 | paper
Whisper, Transformers, Voice Activity Detection, Audio Segmentation, Noisy Recordings

- Contributed dataset processing, baseline evaluation, and cross-domain analysis for WhisperSeg, a Whisper-based VAD system evaluated across human and animal recordings, which reached segment-wise F1 up to 0.9966.
- Used WhisperSeg segmentation outputs in subsequent high-throughput audio workflows.

Large-Scale Audio Clustering with Human-in-the-Loop Review Applied Sciences 2026 | paper | code
Whisper Encoder Embeddings, UMAP, Audio Clustering, Human-in-the-Loop Review, Outlier Analysis

- Designed and evaluated a semi-automated clustering workflow using Whisper encoder embeddings, UMAP, interactive cluster refinement, and outlier correction; achieved ARI = 0.73 and NMI = 0.81 while reducing annotation time by approximately 50% on a 1,454-vocalization benchmark.
- Shifted human effort from item-by-item labeling to cluster definition and outlier review, so annotation effort scales with acoustic diversity rather than dataset size; applied related workflows to 80,000+ hours of recordings.

Technical Skills

Machine Learning: PyTorch, Transformers, VAEs, Flow Matching, Representation Learning, Generative Modeling, Optimal Transport
Speech & Audio ML: Whisper, Foundation Model Adaptation, Voice Activity Detection, Audio Segmentation, Spectrogram Analysis, Acoustic Feature Extraction
Model & Data Evaluation: Audio Model Evaluation, Dataset Construction, Audio Clustering, Human-in-the-Loop Workflows, Outlier Analysis
Software & Infrastructure: Python, NumPy, Pandas, Hugging Face, Git, Linux, HPC, Multi-GPU Training, GPU Batch Processing, Efficient Inference, CTranslate2

Selected Publications

- K. Lee, "Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity in Birdsong," Interspeech 2026, oral presentation. arXiv:2607.03496.
- K. Lee et al., "Benchmarking Automated and Semi-Automated Vocal Clustering Methods," Applied Sciences, 2026.
- N. Gu, K. Lee, et al., "Positive Transfer of the Whisper Speech Transformer to Human and Animal Voice Activity Detection," ICASSP 2024.

Education

ETH Zurich – Ph.D. Candidate, Institute of Neuroinformatics 2022–Present
Audio Machine Learning and Computational Neuroscience; expected completion early 2027
University of Zurich – M.Sc., Neural Systems and Computation, GPA 5.5/6.0 2020–2022
Korea University – B.S. Computer Science; B.Econ. Statistics, GPA 4.02/4.50 2014–2020